

Register Number

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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Computer Science and Engineering**Continuous Assessment Test – I****Regulations – R2021**

Degree & Branch	B.E. & CSE				Semester	VI
Subject Code & Name	UCS2603 & Theory of Computation					
Academic Year	2024-2025 EVEN	Batch	2022-2026	Date	17.02.2025	FN
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1:	Construct mathematical proofs related to computation and finite automata for a given language (K3)
CO2:	Identify relationship between finite automata and regular expressions (K3)
CO3:	Distinguish different types of grammars, analyse relationship between language and context free grammar and construct PDA for CFG (K4)
CO4:	Construct Turing machine for a given language (K3)
CO5:	Explain the decidability or undecidability of various problems and make use of it (K3)

Part – A (4 × 2 = 8 Marks)

		KL	CO	PI
1.	Write the equation for " n^2 " and solve it using Induction on integers.	K3	CO1	1.4.1
2.	Compare DFA with NFA.	K2	CO1	1.4.1
3.	Solve: If $L = \{a^n\}$, where n is divisible by 2 or 3 or $n = 5$, then its corresponding Regular Expression is-----.	K3	CO2	13.3.1
4.	Define CFG with an example.	K2	CO3	1.3.1

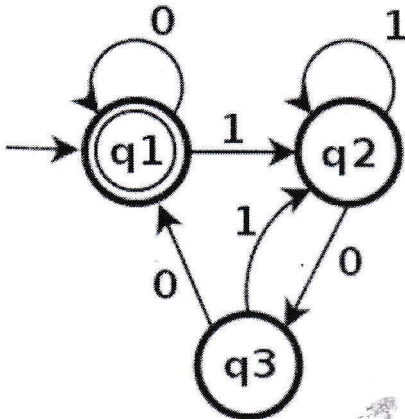
Part – B (3×6 = 18 Marks)

		KL	CO	PI
5.	"For every NFA, there exists a DFA which simulates the behaviour of NFA. If L is the set accepted by NFA, then there exists a DFA which also accepts L ." Explain and write the proof with basis and induction steps.	K2	CO1	2.1.3

6.	Apply <i>Myhill Nerode Theorem</i> to minimize the given DFA and represent the minimized DFA as transition table and transition diagram.	K3	CO2	2.4.1
7.	Explain the <i>Chomsky classification of languages</i> with language name, production rules, examples and its equivalent automata.	K2	CO3	1.4.1

Part – C (2 × 12 = 24 Marks)

		KL	CO	PI																								
8.	(a) Construct a DFA to recognize all strings over $\{x, y\}$ that contain even number of x's and even number of y's. (4 marks) (b) Construct a DFA for the given NEFA M. $M = (\{q_0, q_1, q_2, q_3\}, \{0, 1\}, \delta, q_0, \{q_3\})$, where δ is given as: (8 marks) <table border="1"> <tr> <th rowspan="2">States</th> <th colspan="2">Inputs</th> </tr> <tr> <th>0</th> <th>1</th> </tr> <tr> <td>q_0</td> <td>$\{q_0, q_1\}$</td> <td>$\{q_0\}$</td> </tr> <tr> <td>q_1</td> <td>$\{q_2\}$</td> <td>$\{q_1\}$</td> </tr> <tr> <td>q_2</td> <td>$\{q_3\}$</td> <td>$\{q_3\}$</td> </tr> <tr> <td>q_3</td> <td>ϕ</td> <td>$\{q_2\}$</td> </tr> </table>	States	Inputs		0	1	q_0	$\{q_0, q_1\}$	$\{q_0\}$	q_1	$\{q_2\}$	$\{q_1\}$	q_2	$\{q_3\}$	$\{q_3\}$	q_3	ϕ	$\{q_2\}$	K3	CO1	2.1.3, 2.4.1, 13.3.1, 13.4.1							
States	Inputs																											
	0	1																										
q_0	$\{q_0, q_1\}$	$\{q_0\}$																										
q_1	$\{q_2\}$	$\{q_1\}$																										
q_2	$\{q_3\}$	$\{q_3\}$																										
q_3	ϕ	$\{q_2\}$																										
(Or)																												
9.	(a) Construct the NFA from the given ϵ-NFA and write the result as transition table. (8 marks) <table border="1"> <tr> <th>States</th> <th>ϵ</th> <th>0</th> <th>1</th> </tr> <tr> <td>$\rightarrow A$</td> <td>$\{B, D\}$</td> <td>$\{A\}$</td> <td>$\emptyset$</td> </tr> <tr> <td>B</td> <td>$\emptyset$</td> <td>$\{C\}$</td> <td>$\{E\}$</td> </tr> <tr> <td>C</td> <td>$\emptyset$</td> <td>$\emptyset$</td> <td>$\{B\}$</td> </tr> <tr> <td>D</td> <td>$\emptyset$</td> <td>$\{E\}$</td> <td>$\{D\}$</td> </tr> <tr> <td>$*E$</td> <td>$\emptyset$</td> <td>$\emptyset$</td> <td>$\emptyset$</td> </tr> </table>	States	ϵ	0	1	$\rightarrow A$	$\{B, D\}$	$\{A\}$	$\emptyset$	B	$\emptyset$	$\{C\}$	$\{E\}$	C	$\emptyset$	$\emptyset$	$\{B\}$	D	$\emptyset$	$\{E\}$	$\{D\}$	$*E$	$\emptyset$	$\emptyset$	$\emptyset$	K3	CO1	2.1.3, 2.4.1, 13.3.1, 13.4.1
States	ϵ	0	1																									
$\rightarrow A$	$\{B, D\}$	$\{A\}$	$\emptyset$																									
B	$\emptyset$	$\{C\}$	$\{E\}$																									
C	$\emptyset$	$\emptyset$	$\{B\}$																									
D	$\emptyset$	$\{E\}$	$\{D\}$																									
$*E$	$\emptyset$	$\emptyset$	$\emptyset$																									

	(b) Infer the language of the converted NFA and justify your answer by checking at least one valid and one invalid strings from the language of NFA as per the transition rule property. (4 marks)			
10.	(a) Construct the ϵ -NFA of RE= $(0+11) 0^*+1^*$ (4 marks) (b) Infer the equivalent DFA of the constructed ϵ -NFA (10.a) using subset construction with lazy evaluation. (8 marks)	K3	CO2	2.1.3, 2.4.1, 13.3.1, 13.4.1
(Or)				
11.	Consider the following DFA D  (a) Write the equations for basis and induction steps ($K=0$ & $K \geq 1$) (4 marks) (b) Apply the equation method and give all the regular expressions $R_{ij}^{(0)}$ and $R_{ij}^{(1)}$. Find $R_{11}^{(2)}$ using the prior $R_{ij}^{(k)}$ values. Try to simplify the expressions as much as possible. (8 marks)	K3	CO2	2.1.3, 2.4.1, 13.3.1, 13.4.1

Prepared By	Reviewed By	Approved By
Course Coordinator	PAC Team	HOD

Answer Key

Part A

1. Induction on integer for n^2

Prove by induction :

$$\sum_{i=0}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

Solution :

$$\text{Let } P(n) : 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Basis step:

For $n = 1$

L.H.S. = 1

$$\text{R.H.S} = \frac{1(2)(3)}{6} = 1$$

Inductive step:

Assume the result for $n = k$

$$1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

We have to prove the result for $n = k + 1$

$$\begin{aligned} 1^2 + 2^2 + \dots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= (k+1) \left[\frac{k(2k+1)}{6} + (k+1) \right] \\ &= (k+1) \left[\frac{2k^2 + k + 6k + 6}{6} \right] \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\ &= (k+1) \frac{(2k^2 + 4k + 3k + 6)}{6} \\ &= (k+1) \frac{(2k(k+2) + 3(k+2))}{6} \\ &= \frac{(k+1)(k+2) \cdot (2k+3)}{6} \end{aligned}$$

2. NFA Vs DFA

DFA	NFA
For each symbol, a DFA transitions to a single definite state & it should be defined for all states.	An NFA can transition to zero, one, or multiple states & may not be defined for all states.
DFA does not allow transitions without consuming input symbols.	NFA allows transitions without consuming input symbols (ϵ -transitions).
Example	Example

3. n is divisible by 2 or 3 or $n = 5$

$$RE = (aa)^* + (aaa)^* + aaaaa$$

4. CFG with Example

$G = (V, T, P, S)$ Explanation and suitable example

Part B

5. For every NFA, there exists a DFA which simulates the behaviour of NFA. If L is the set accepted by NFA, then there exists a DFA which also accepts L .

Let $M = (Q, \Sigma, q_0, F, \delta)$ be NFA accepting L

we construct DFA $M^1 = (Q^1, \Sigma, q_0^1, F^1, \delta^1)$, where
explain the parameters of M^1 .

Basis

The result is trivial if string length is 0 i.e., $|x| = 0$

since $q_0^1 = [q_0]$. x must be ϵ

Induction

Suppose the hypothesis is true for inputs of length m .

Let xa be a string of length $m+1$ with a in Σ .

$$\text{Then } \delta^1(q_0^1, xa) = \delta^1(\delta^1(q_0^1, x), a)$$

By induction hypothesis

$$\delta^1(q_0^1, x) = [p_1, p_2, \dots, p_j]$$

if and only if

$$\delta(q_0, x) = \{p_1, p_2, \dots, p_j\}$$

By definition of δ^1

$$\delta^1([p_1, p_2, \dots, p_j], a) = [r_1, r_2, \dots, r_k]$$

if and only if

$$\delta(\{p_1, p_2, \dots, p_j\}, a) = \{r_1, r_2, \dots, r_k\}$$

Thus

$$\delta^1(q_0^1, xa) = [r_1, r_2, \dots, r_k]$$

if and only if

$$\delta(q_0, xa) = \{r_1, r_2, \dots, r_k\}$$

which establishes the inductive hypothesis.

Thus $L(M) = L(M^1)$

6. Myhill Nerode Theorem to minimize the given DFA

Reference: <https://www.geeksforgeeks.org/minimization-of-dfa-using-myhill-nerode-theorem/>

Transition table

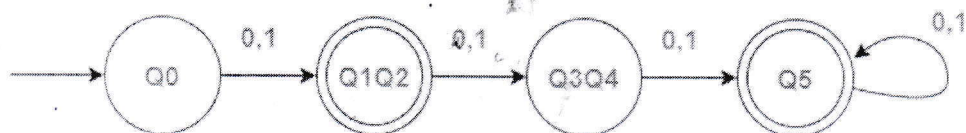
States	Inputs	
	0	1
Q0	Q1	Q2
Q1	Q3	Q4
Q2	Q4	Q3
Q3	Q5	Q5
Q4	Q5	Q5
Q5	Q5	Q5

Table filling

	Q0	Q1	Q2	Q3	Q4	Q5
Q0						
Q1	✓					
Q2	✓					
Q3	✓	✓	✓			
Q4	✓	✓	✓			
Q5	✓	✓	✓	✓	✓	

Transition table & diagram for Minimized DFA

States	Inputs	
	0	1
Q0	Q1Q2	Q1Q2
Q1Q2	Q3Q4	Q3Q4
Q3Q4	Q5	Q5
Q5	Q5	Q5



7. *Chomsky classification of languages* with language name, production rules, examples and its equivalent automata.

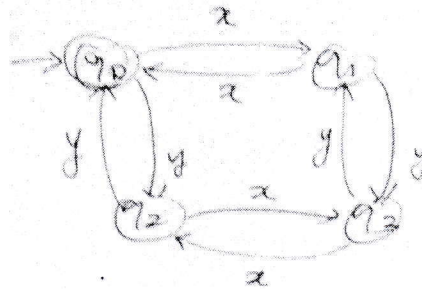
(CPTO)

Type	Grammar / Language	Form of production in grammar	Example	Automata (or) Accepting device
0	Unrestricted grammar (or) Phrase structure grammar	A grammar without any restrictions, with the following production form : $\phi A \psi \rightarrow \phi \alpha \psi$ Λ - variable ϕ - left context ψ - right context α - replacement string	(i) $ab\Lambda bcd \rightarrow abABbcd$ where ab -left context bcd -right context $\alpha = AB$ (ii) $\Lambda C \rightarrow A$ where A - left context λ - right context $\alpha = \lambda$ (iii) $C \rightarrow \lambda$ λ - left and right context $\alpha = \lambda$	Turing Machine (TM)
1	Context sensitive grammar (or) Context dependent grammar	A production of the form $\phi A \psi \rightarrow \phi \alpha \psi$ is called type 1 production if $\alpha \neq \lambda$. In type 1 productions erasing of A is not permitted.	(i) $a\Lambda bcd \rightarrow abcDbcd$ where a - left context bcd - right context Λ is replaced by $bcd \neq \lambda$ (ii) $\Lambda B \rightarrow A\lambda BC$ where A - left context λ - right context $\alpha = bBC \neq \lambda$	Linear Bounded Automata (LBA)

			(iii) $A \rightarrow abA$ where λ - left & right context, $A = abA \neq \lambda$ (iv) $A \rightarrow \lambda$ is allowed but A does not appear on the right handside of any production of the grammar.	
2	Context free grammar	A production of the form $A \rightarrow \alpha$, where $A \in V_N$ and $\alpha \in (V_N \cup \Sigma)^*$ is called a type 2 grammar. In simple words left hand side has no left context or right context. $A = \text{L.H.S.} \in V_N$ $\alpha = \text{R.H.S.} \in (V_N \cup \Sigma)^*$	(i) $S \rightarrow Aa$ (ii) $A \rightarrow a$ (iii) $B \rightarrow abc$ (iv) $A \rightarrow \lambda$ where L.H.S. $\in V_N$ (Variables) and R.H.S. $\in (V_N \cup \Sigma)^*$ (Combination of variable & terminal)	Push Down Automata (PDA)
3	Regular grammar	A production of the form $A \rightarrow a$ or $A \rightarrow aB$ where $A, B \in V_N$ and $a \in \Sigma$ is called a type 3 production. A production $S \rightarrow \lambda$ is allowed in type 3, but in this case S does not appear on the right hand side of any production. $A = \text{L.H.S.} \in V_N$ $\text{R.H.S.} \in (V_N \cup a)$	(i) $S \rightarrow b \mid c$ (ii) $S \rightarrow bA$ (iii) $S \rightarrow a$ where L.H.S. $\in V_N$ (Variables) and R.H.S. belongs to either a single terminal or a terminal followed by one variable.	Finite Automata (FA)

Part C

8. (a) Construct a DFA to recognize all strings over $\{x, y\}$ that contain even number of x 's and even number of y 's.



- (b) Construct a DFA for the given NFA M.

$M = (\{q_0, q_1, q_2, q_3\}, \{0, 1\}, \delta, q_0, \{q_3\})$, where δ is given as: (8 marks)

States	Inputs	
	0	1
q_0	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_1\}$
q_2	$\{q_3\}$	$\{q_3\}$
q_3	ϕ	$\{q_2\}$

Solution:

DFA transition table is :

States	Inputs	
	a	b
$[q_0]$	$[q_0, q_1]$	$[q_0]$
$[q_0, q_1]$	$[q_0, q_1, q_2]$	$[q_0, q_1]$
$[q_0, q_1, q_2]$	$[q_0, q_1, q_2, q_3]$	$[q_0, q_1, q_3]$
$[q_0, q_1, q_3]$	$[q_0, q_1, q_2]$	$[q_0, q_1, q_2]$
$[q_0, q_1, q_2, q_3]$	$[q_0, q_1, q_2, q_3]$	$[q_0, q_1, q_2, q_3]$

The final states are

$$F^1 = [q_0, q_1, q_3] \text{ and } [q_0, q_1, q_2, q_3]$$

(11)

$$\begin{aligned} \delta'(B, 0) &= \hat{\delta}(B, 0) = \text{Eclose}(\delta(\hat{\delta}(B, \epsilon), 0)) \\ &= \eta(\delta(B, 0)) \\ &= \eta(C) = \{C\}. \end{aligned}$$

$$\begin{aligned} \delta'(B, 1) &= \hat{\delta}(B, 1) = \text{Eclose}(\delta(\hat{\delta}(B, \epsilon), 1)) \\ &= \eta(\delta(B, 1)) \\ &= \eta(E) = \{E\}. \end{aligned}$$

$$\begin{aligned} \delta'(C, 0) &= \hat{\delta}(C, 0) = \text{Eclose}(\delta(\hat{\delta}(C, \epsilon), 0)) \\ &= \eta(\delta(C, 0)) \\ &= \emptyset \end{aligned}$$

$$\begin{aligned} \delta'(C, 1) &= \hat{\delta}(C, 1) = \text{Eclose}(\delta(\hat{\delta}(C, \epsilon), 1)) \\ &= \eta(\delta(C, 1)) \\ &= \eta(B) = \{B\} \end{aligned}$$

$$S'(A,0) = \hat{S}(A,0) = \xi \text{ clos } (S(\hat{S}(A,\xi),0))$$

$$= \eta(S(A,0))$$

$$= \eta(E) = \{E\}$$

$$S'(A,1) = \hat{S}(A,1) = \xi \text{ clos } (S(\hat{S}(A,\xi),1))$$

$$= \eta(S(A,1))$$

$$= \eta(A) = \{A\}$$

$$S'(E,0) = \hat{S}(E,0) = \xi \text{ clos } (S(\hat{S}(E,\xi),0))$$

$$= \eta(S(E,0))$$

$$= \emptyset$$

$$S'(E,1) = \hat{S}(E,1) = \xi \text{ clos } (S(\hat{S}(E,\xi),1))$$

$$= \eta(S(E,1))$$

$$= \emptyset$$

$\rightarrow A$	$\{A, B, C, D, E\}$	$\{A, E\}$
B	$\{C\}$	$\{E\}$
C	$\emptyset$	$\{B\}$
D	$\{E\}$	$\{D\}$
E	$\emptyset$	$\emptyset$

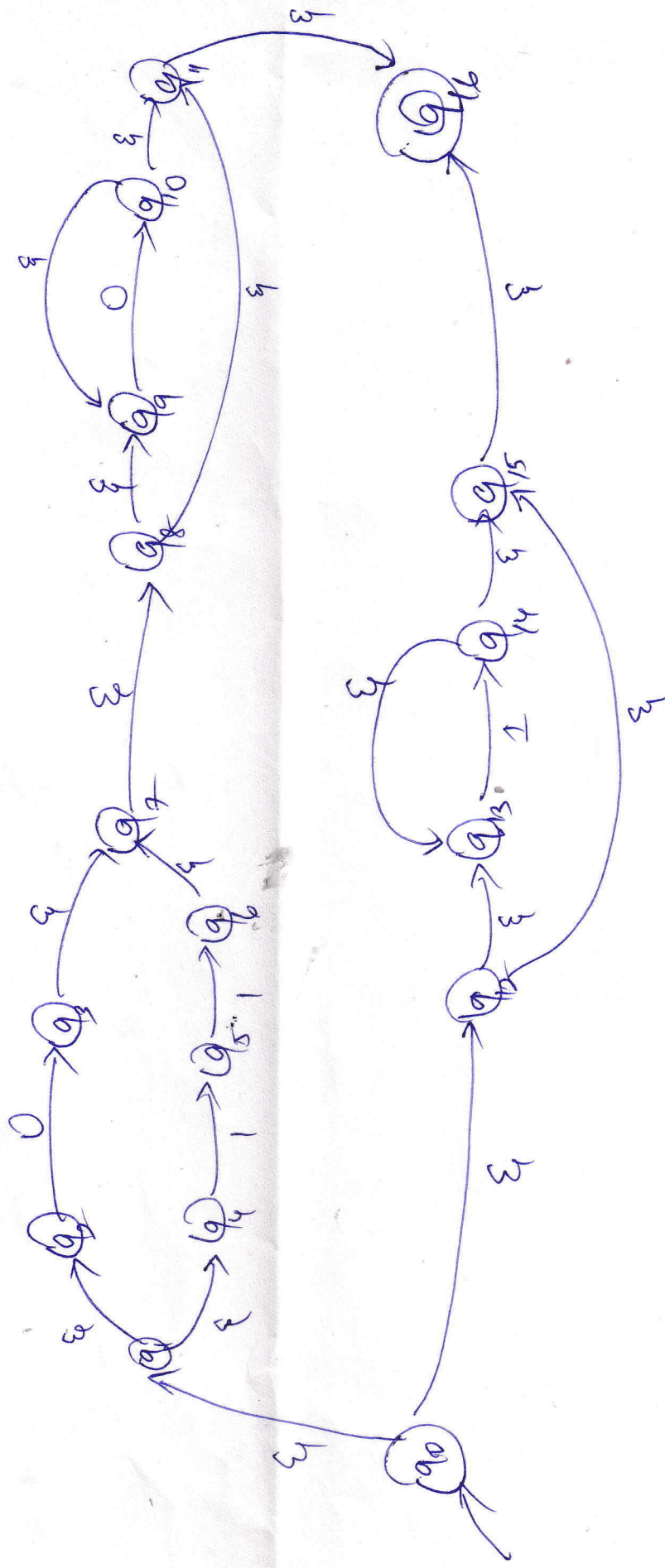
(10) (a)

$$RE = \underline{\underline{(0+1)^* \cdot 0^* + 1^*}}$$

Order of construction $(0+1)^*$, 0^* , 1^*

$$(0+1)^* \cdot 0^*$$

$$(0+1)^* \cdot 0^* + 1^*$$



$$\begin{aligned}\xi_{\text{closure}}(A_0) &= \{q_0, q_1, q_{12}, q_2, q_4, q_{15}, q_{16}, \\ &= \{q_0, q_1, q_2, q_4, q_{12}, q_{15}, q_{16}\} - (A)\end{aligned}$$

$$\text{move}(A, 0) = \{q_3\}$$

$$\xi_{\text{closure}}[\text{move}(A, 0)] = \{q_3, q_7, q_8, q_9, q_{14}, q_{16}\} - (B)$$

$$\text{move}(A, 1) = \{q_5, q_{14}\}$$

$$\begin{aligned}\xi_{\text{closure}}[\text{move}(A, 1)] &= \{q_5, q_{14}, q_{15}, q_{16}, q_{13}\} \\ &= \{q_5, q_{13}, q_{14}, q_{15}, q_{16}\} - (C)\end{aligned}$$

$$\text{move}(B, 0) = \{q_{10}\}$$

$$\begin{aligned}\xi_{\text{closure}}[\text{move}(B, 0)] &= \{q_{10}, q_9, q_{11}, q_{16}\} \\ &= \{q_9, q_{10}, q_{11}, q_{16}\} - (D)\end{aligned}$$

$$\text{move}(B, 1) = \{\varnothing\}$$

$$\xi_{\text{closure}}[\text{move}(B, 1)] = \varnothing$$

$$\text{move}(C, 0) = \{\varnothing\}$$

$$\xi_{\text{closure}}[\text{move}(C, 0)] = \varnothing$$

$$\text{move}(C, 1) = \{q_{16}, q_{14}\}$$

$$\begin{aligned}\xi_{\text{closure}}[\text{move}(C, 1)] &= \{q_{16}, q_{14}, q_7, q_8, q_9, q_{11}, q_{16}\} \\ &= \{q_7, q_8, q_9, q_{11}, q_{14}, q_{16}\} - (E)\end{aligned}$$

$$\text{move}(A, 0) = \{q_{10}\}$$

$$\xi\text{closure}[\text{move}(A, 0)] = D$$

$$\text{move}(A, 1) = \phi$$

$$\xi\text{closure}[\text{move}(A, 1)] = \phi$$

$$\text{move}(B, 0) = \{q_{10}\}$$

$$\xi\text{closure}[\text{move}(B, 0)] = D$$

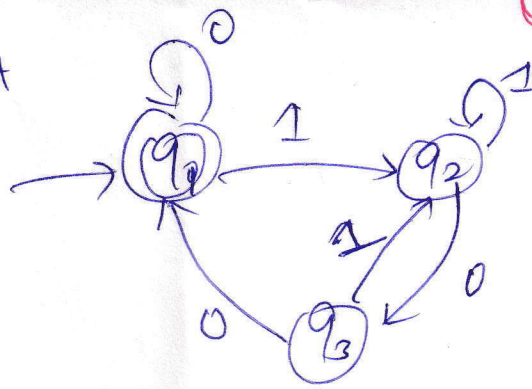
$$\text{move}(E, 1) = \phi$$

$$\xi\text{closure}[\text{move}(E, 1)] = \phi$$

DFA:

States	0	1
$\rightarrow A$	B	C
B	D	ϕ
C	ϕ	E
D	D	ϕ
E	D	ϕ

(1) Given DFA



(a) Basis step: $\boxed{i=j}$ $R_{ij}^{(0)} = R_{ii}^{(0)} = \{a \mid \delta(q_i, a) = q_i\} \cup \{\epsilon\}$
 $= a + \epsilon.$

$\boxed{i \neq j}$ $R_{ij}^{(0)} = \{a \mid \delta(q_i, a) = q_j\}$
 $= a$

Induction:

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} \cdot (R_{kk}^{(k-1)})^* \cdot R_{kj}^{(k-1)}$$

(b) $k=0$

$R_{11}^{(0)} = (0 + \epsilon)$	$R_{21}^{(0)} = \emptyset$	$R_{31}^{(0)} = \emptyset$
$R_{12}^{(0)} = 1$	$R_{22}^{(0)} = (1 + \epsilon)$	$R_{32}^{(0)} = 1$
$R_{13}^{(0)} = \emptyset$	$R_{23}^{(0)} = \emptyset$	$R_{33}^{(0)} = \epsilon$

$k=1$ $R_{ij}^{(1)} = R_{ij}^{(0)} + R_{ik}^{(0)} \cdot (R_{kk}^{(0)})^* \cdot R_{kj}^{(0)}$

$$\begin{aligned}
 R_{11}^{(1)} &= (0 + \epsilon) + (0 + \epsilon) \cdot (0 + \epsilon)^* \cdot (0 + \epsilon) \\
 &= (0 + \epsilon) [\epsilon + (0 + \epsilon)^* (0 + \epsilon)] \\
 &= (0 + \epsilon) (0 + \epsilon)^*
 \end{aligned}$$

$$\begin{aligned}
 R_{12}^{(1)} &= 1 + (0+\xi)(0+\xi)^* (1) \\
 &= [\xi + (0+\xi)(0+\xi)^*] 1 \\
 &= (0+\xi)^* \cdot 1
 \end{aligned}$$

$$R_{13}^{(1)} = \phi + \phi = \phi$$

$$R_{21}^{(1)} = \phi + \phi = \phi$$

$$R_{22}^{(1)} = (1+\xi) + \phi \dots = (1+\xi)$$

$$R_{23}^{(1)} = 0 + \phi \dots = 0$$

$$\begin{aligned}
 R_{31}^{(1)} &= 0 + 0 (0+\xi)^* (0+\xi) \\
 &= 0 [\xi + (0+\xi)^* (0+\xi)] \\
 &= 0 (0+\xi)^*
 \end{aligned}$$

$$\begin{aligned}
 R_{32}^{(1)} &= 1 + 0 (0+\xi)^* 1 \\
 &= [\xi + 0 (0+\xi)^*] 1 \\
 &= [\xi + 0 0^*] 1 \\
 &= 0^* 1
 \end{aligned}$$

$$R_{33}^{(1)} = \xi + \underline{0 (0+\xi)^* \phi} = \xi$$

$$\begin{aligned}
 R_{11}^{(2)} &= R_{11}^{(1)} + R_{12}^{(1)} \cdot (R_{22}^{(1)})^* \cdot R_{21}^{(1)} \\
 &= (0+\xi)(0+\xi)^* + \underline{(0+\xi)^* \cdot 1 \cdot (1+\xi)^* \phi}
 \end{aligned}$$

$$\boxed{R_{11}^{(2)} = 0 0^*}$$